Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Double Creek Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
1320 S Arboleda DR	Merced	Merced	95340

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0067-0030-0015-0000

B. OPERATORS

Strickland, Matt

Operator name: Strickland, Matt

Telephone no.: Landline: ; Cellular: (209) 769-2813

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2128	Merced	CA	95344

This operator is responsible for paying permit fees.

C. OWNERS

Strickland, Robert & Victoria

Legal owner name: Strickland, Robert & Victoria

Telephone no.: Landline: (209) 394-3600; Cellular: (209) 769-1256

Mailing Address Number and Street	City	State	Zip Code
8632 Meadow DR	Winton	CA	95388

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	175	769	736	564	402
Number under roof	1,650	0	0	0	0	0
Maximum number	1,680	200	800	750	580	450
Average number	1,650	175	770	740	560	400
Avg live weight (lbs)	1,400	1,500	900	700		

Predominant milk cow breed: Holstein

Average milk production: 80 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 64,996.68 *tons per reporting period*

Total nitrogen from manure: 802,988.25 *lbs per reporting period*

After ammonia losses (30% loss applied): 562,091.78 *lbs per reporting period*

Total phosphorus from manure: 130,274.13 *lbs per reporting period*

Total potassium from manure: 326,830.72 *lbs per reporting period*

Total salt from manure: 817,143.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 29,447,620 *gallons*

Total nitrogen generated: 219,235.93 *lbs*

Total phosphorus generated: 17,046.25 *lbs*

Total potassium generated: 222,987.63 *lbs*

Total salt generated: 749,536.29 *lbs*

	29,447,620	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	29,447,620	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CSR Well	Ground water
DCR Well	Ground water
MID	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	Corral solids	640.00 *ton*	Dry-weight	16.3		20,100.00	6,800.00	42,000.00		0.00
10/20/2023	Corral solids	2,790.00 *ton*	Dry-weight	16.3		20,100.00	6,800.00	42,000.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	115,410.58	39,044.38	241,156.44	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	115,410.58	39,044.38	241,156.44	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
CS 1	69	69	2	process wastewater	0053-0120-0026-0000
CS 2	73	73	2	process wastewater	0053-0120-0024-0000
CS 3	71	71	2	process wastewater	0053-0120-0027-0000
DCR 1A	41	41	2	process wastewater	0067-0030-0015-0000
DCR 1B	16	16	2	process wastewater	0067-0030-0015-0000
DCR 1C	36	36	2	process wastewater	0067-0030-0015-0000
DCR 1D	29	29	2	process wastewater	0067-0030-0015-0000
DCR 2A	26	26	2	process wastewater	0067-0030-0015-0000
DCR 2B	30	30	2	process wastewater	0067-0030-0015-0000
DCR 2C	18	18	2	process wastewater	0067-0030-0015-0000
DCR 2D	29	29	2	process wastewater	0067-0030-0015-0000
OC 1	36	36	2	process wastewater	0067-0030-0024-0000
Totals for areas that were used for application	474	474	24		
Totals for areas that were not used for application					
Land application area totals	474	474	24		

B. CROPS AND HARVESTS

CS 1

Field name: CS 1

11/16/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 69 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	716.56 *ton*	As-is		54.5	7,740.00	910.00	6,100.00		8.49

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	10.38	160.76	18.90	126.70	802.33

CS 1

06/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 69 Plant date: 06/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	1,632.97 *ton*	As-is		63.1	4,320.00	1,000.00	6,010.00		3.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.67	204.48	47.33	284.47	541.44

CS 2

Field name: CS 2

11/16/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 73 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	757.92 *ton*	As-is		63.8	6,370.00	690.00	8,330.00		8.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	10.38	132.27	14.33	172.97	668.25

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 73 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	2,172.24 *ton*	As-is		63.5	5,260.00	700.00	6,470.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.76	313.04	41.66	385.05	1,737.79

CS 3

Field name: CS 3

11/16/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 71 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	1,012.45 *ton*	As-is		61.7	5,360.00	860.00	5,940.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	14.26	152.87	24.53	169.41	1,103.23

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 71 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,680.76 *ton*	As-is		66.8	3,810.00	510.00	5,370.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.67	180.39	24.15	254.24	1,131.74

DCR 1A

Field name: DCR 1A

11/15/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 36 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	581.81 *ton*	As-is		57.2	7,990.00	800.00	5,620.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	16.16	258.26	25.86	181.65	1,687.77

DCR 1A

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	965.85 *ton*	As-is		61.9	4,530.00	1,300.00	5,680.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.83	243.07	69.76	304.78	960.86

DCR 1B

Field name: DCR 1B

11/15/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 16 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	226.88 *ton*	As-is		45.8	10,200.00	900.00	6,630.00	100.00	8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	14.18	289.27	25.52	188.03	1,368.03

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	376.64 *ton*	As-is		61.9	4,760.00	1,560.00	6,320.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.54	224.10	73.44	297.55	968.62

DCR 1C

Field name: DCR 1C

11/15/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 36 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	510.48 *ton*	As-is		50.2	9,190.00	850.00	5,810.00	100.00	10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	14.18	260.63	24.11	164.77	1,497.07

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	847.44 *ton*	As-is		64.0	4,680.00	1,330.00	5,540.00		4.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.54	220.33	62.62	260.82	711.85

DCR 1D

Field name: DCR 1D

11/15/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 29 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	413.91 *ton*	As-is		53.5	6,650.00	950.00	6,140.00	100.00	9.34

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	14.27	189.83	27.12	175.27	1,239.76

DCR 1D

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	687.13 *ton*	As-is		63.8	5,070.00	1,190.00	5,320.00		3.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.69	240.26	56.39	252.11	514.64

DCR 2A

Field name: DCR 2A

11/15/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 26 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	363.15 *ton*	As-is		60.0	5,000.00	980.00	6,760.00	100.00	10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	13.97	139.67	27.38	188.84	1,150.91

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	695.82 *ton*	As-is		64.2	4,330.00	1,120.00	6,760.00	100.00	5.27

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.76	231.76	59.95	361.83	1,009.83

DCR 2B

Field name: DCR 2B

11/15/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 30 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	418.31 *ton*	As-is		57.7	6,130.00	920.00	6,650.00	100.00	9.54

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	13.94	170.95	25.66	185.45	1,125.37

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	801.52 *ton*	As-is		52.4	5,570.00	1,400.00	9,760.00	100.00	5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.72	297.63	74.81	521.52	1,424.35

DCR 2C

Field name: DCR 2C

11/18/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 18 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	257.51 *ton*	As-is		65.1	4,750.00	920.00	5,180.00	100.00	7.78

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	14.31	135.91	26.32	148.21	776.88

DCR 2C

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	493.41 *ton*	As-is		61.8	4,350.00	1,110.00	7,210.00	100.00	5.93

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.41	238.48	60.85	395.28	1,241.89

DCR 2D

Field name: DCR 2D

11/18/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 29 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	412.64 *ton*	As-is		53.0	8,180.00	1,170.00	7,980.00	100.00	10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	14.23	232.79	33.30	227.09	1,431.15

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	790.65 *ton*	As-is		64.1	4,560.00	1,080.00	6,180.00	100.00	5.03

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.26	248.65	58.89	336.98	984.64

OC 1

Field name: OC 1

11/18/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 36 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	510.48 *ton*	As-is		64.0	5,440.00	810.00	11,100.00		9.98

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	10.00	110.00	17.00	83.00	0.00
Total actual harvest content	14.18	154.28	22.97	314.80	1,018.92

07/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 07/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	515.82 *ton*	As-is		59.3	5,540.00	650.00	6,050.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	14.33	158.76	18.63	173.37	863.08

NUTRIENT BUDGET

A. LAND APPLICATIONS

CS 1 - 11/16/2022: Wheat, silage, soft dough

Field name: CS 1

Crop: Wheat, silage, soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	178.85	9.92	139.88	636.78	1,438,560.00 *gal*
Application event totals		178.85	9.92	139.88	636.78	

CS 1 - 06/21/2023: Corn, silage

Field name: CS 1

Crop: Corn, silage

Plant date: 06/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.28	0.00	0.00	22.44	18,552,323.40 *gal*
Application event totals		6.28	0.00	0.00	22.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.28	0.00	0.00	22.44	18,552,323.40 *gal*
Application event totals		6.28	0.00	0.00	22.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.28	0.00	0.00	22.44	18,552,323.40 *gal*
Application event totals		6.28	0.00	0.00	22.44	

CS 1 - 06/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	239.68	15.90	128.98	777.17	2,700,000.00 *gal*
MID	Surface water	4.48	0.00	0.00	16.00	13,229,874.10 *gal*
Application event totals		244.16	15.90	128.98	793.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.28	0.00	0.00	22.44	18,552,323.40 *gal*
Application event totals		6.28	0.00	0.00	22.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.28	0.00	0.00	22.44	18,552,323.40 *gal*
Application event totals		6.28	0.00	0.00	22.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.28	0.00	0.00	22.44	18,552,323.40 *gal*
Application event totals		6.28	0.00	0.00	22.44	

CS 2 - 11/16/2022: Wheat, silage, soft dough

Field name: CS 2

Crop: Wheat, silage, soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	169.05	9.37	132.22	601.88	1,438,560.00 *gal*
Application event totals		169.05	9.37	132.22	601.88	

CS 2 - 06/08/2023: Corn, silage

Field name: CS 2

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	364.45	42.96	677.80	1,259.29	3,240,000.00 *gal*
MID	Surface water	6.28	0.00	0.00	22.43	19,623,069.60 *gal*
Application event totals		370.74	42.96	677.80	1,281.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.28	0.00	0.00	22.43	19,623,069.60 *gal*
Application event totals		6.28	0.00	0.00	22.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.28	0.00	0.00	22.43	19,623,069.60 *gal*
Application event totals		6.28	0.00	0.00	22.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.28	0.00	0.00	22.43	19,623,069.60 *gal*
Application event totals		6.28	0.00	0.00	22.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.28	0.00	0.00	22.43	19,623,069.60 *gal*
Application event totals		6.28	0.00	0.00	22.43	

CS 2 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.28	0.00	0.00	22.43	19,623,069.60 *gal*
Application event totals		6.28	0.00	0.00	22.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.28	0.00	0.00	22.43	19,623,069.60 *gal*
Application event totals		6.28	0.00	0.00	22.43	

CS 3 - 11/16/2022: Wheat, silage, soft dough

Field name: CS 3

Crop: Wheat, silage, soft dough Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	173.82	9.64	135.94	618.84	1,438,560.00 *gal*
CSR Well	Ground water	0.90	0.00	0.00	0.50	7,128,000.00 *gal*
Application event totals		174.71	9.64	135.94	619.34	

CS 3 - 06/26/2023: Corn, silage

Field name: CS 3

Crop: Corn, silage Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

CS 3 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.29	0.00	0.00	22.47	19,114,416.00 *gal*
Application event totals		6.29	0.00	0.00	22.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.29	0.00	0.00	22.47	19,114,416.00 *gal*
Application event totals		6.29	0.00	0.00	22.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.29	0.00	0.00	22.47	19,114,416.00 *gal*
Application event totals		6.29	0.00	0.00	22.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.29	0.00	0.00	22.47	19,114,416.00 *gal*
Application event totals		6.29	0.00	0.00	22.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.29	0.00	0.00	22.47	19,114,416.00 *gal*
Application event totals		6.29	0.00	0.00	22.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	155.29	10.30	83.57	503.52	1,800,000.00 *gal*
MID	Surface water	6.29	0.00	0.00	22.47	19,114,416.00 *gal*
Application event totals		161.58	10.30	83.57	525.99	

CS 3 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.29	0.00	0.00	22.47	19,114,416.00 *gal*
Application event totals		6.29	0.00	0.00	22.47	

DCR 1A - 11/15/2022: Wheat, silage, soft dough

Field name: DCR 1A

Crop: Wheat, silage, soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	287.40	33.88	534.50	993.06	1,260,000.00 *gal*
Application event totals		287.40	33.88	534.50	993.06	

DCR 1A - 06/24/2023: Corn, silage

Field name: DCR 1A

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	246.34	29.04	458.14	851.19	1,080,000.00 *gal*
MID	Surface water	7.33	0.00	0.00	26.19	11,297,252.00 *gal*
Application event totals		253.68	29.04	458.14	877.38	

DCR 1A - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	7.33	0.00	0.00	26.19	11,297,252.00 *gal*
Application event totals		7.33	0.00	0.00	26.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	7.33	0.00	0.00	26.19	11,297,252.00 *gal*
Application event totals		7.33	0.00	0.00	26.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	7.33	0.00	0.00	26.19	11,297,252.00 *gal*
Application event totals		7.33	0.00	0.00	26.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	7.33	0.00	0.00	26.19	11,297,252.00 *gal*
Application event totals		7.33	0.00	0.00	26.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	7.33	0.00	0.00	26.19	11,297,252.00 *gal*
Application event totals		7.33	0.00	0.00	26.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	7.33	0.00	0.00	26.19	11,297,252.00 *gal*
Application event totals		7.33	0.00	0.00	26.19	

DCR 1B - 11/15/2022: Wheat, silage, soft dough

DCR 1B - 11/15/2022: Wheat, silage, soft dough

Field name: DCR 1B

Crop: Wheat, silage, soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	346.42	40.84	644.26	1,196.99	675,000.00 *gal*
Application event totals		346.42	40.84	644.26	1,196.99	

DCR 1B - 06/22/2023: Corn, silage

Field name: DCR 1B

Crop: Corn, silage

Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.43	0.00	0.00	22.98	4,405,504.87 *gal*
Application event totals		6.43	0.00	0.00	22.98	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.43	0.00	0.00	22.98	4,405,504.80 *gal*
Application event totals		6.43	0.00	0.00	22.98	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.43	0.00	0.00	22.98	4,405,504.80 *gal*
Application event totals		6.43	0.00	0.00	22.98	

DCR 1B - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.43	0.00	0.00	22.98	4,405,504.80 *gal*
Application event totals		6.43	0.00	0.00	22.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.43	0.00	0.00	22.98	4,405,504.80 *gal*
Application event totals		6.43	0.00	0.00	22.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.43	0.00	0.00	22.98	4,405,504.80 *gal*
Application event totals		6.43	0.00	0.00	22.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	206.73	13.72	111.25	670.31	540,000.00 *gal*
MID	Surface water	6.43	0.00	0.00	22.98	4,405,504.80 *gal*
Application event totals		213.16	13.72	111.25	693.29	

DCR 1C - 11/15/2022: Wheat, silage, soft dough

Field name: DCR 1C

Crop: Wheat, silage, soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	312.76	17.34	244.61	1,113.54	1,312,500.00 *gal*
Application event totals		312.76	17.34	244.61	1,113.54	

DCR 1C - 06/23/2023: Corn, silage

Field name: DCR 1C

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.43	0.00	0.00	22.98	9,912,385.70 *gal*
Application event totals		6.43	0.00	0.00	22.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	260.71	17.30	140.30	845.34	1,532,250.00 *gal*
MID	Surface water	6.43	0.00	0.00	22.98	9,912,385.70 *gal*
Application event totals		267.14	17.30	140.30	868.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.43	0.00	0.00	22.98	9,912,385.70 *gal*
Application event totals		6.43	0.00	0.00	22.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.43	0.00	0.00	22.98	9,912,385.70 *gal*
Application event totals		6.43	0.00	0.00	22.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.43	0.00	0.00	22.98	9,912,385.70 *gal*
Application event totals		6.43	0.00	0.00	22.98	

DCR 1C - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.43	0.00	0.00	22.98	9,912,385.70 *gal*
Application event totals		6.43	0.00	0.00	22.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.43	0.00	0.00	22.98	9,912,385.70 *gal*
Application event totals		6.43	0.00	0.00	22.98	

DCR 1D - 11/15/2022: Wheat, silage, soft dough

Field name: DCR 1D

Crop: Wheat, silage, soft dough Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	252.92	14.02	197.81	900.48	855,000.00 *gal*
Application event totals		252.92	14.02	197.81	900.48	

DCR 1D - 06/24/2023: Corn, silage

Field name: DCR 1D

Crop: Corn, silage Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

DCR 1D - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.48	0.00	0.00	23.13	8,037,439.40 *gal*
Application event totals		6.48	0.00	0.00	23.13	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.48	0.00	0.00	23.13	8,037,439.40 *gal*
Application event totals		6.48	0.00	0.00	23.13	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.48	0.00	0.00	23.13	8,037,439.40 *gal*
Application event totals		6.48	0.00	0.00	23.13	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.48	0.00	0.00	23.13	8,037,439.40 *gal*
Application event totals		6.48	0.00	0.00	23.13	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	228.11	15.13	122.76	739.65	1,080,000.00 *gal*
MID	Surface water	6.48	0.00	0.00	23.13	8,037,439.40 *gal*
Application event totals		234.59	15.13	122.76	762.78	

09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.48	0.00	0.00	23.13	8,037,439.40 *gal*
Application event totals		6.48	0.00	0.00	23.13	

DCR 1D - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.48	0.00	0.00	23.13	8,037,439.40 *gal*
Application event totals		6.48	0.00	0.00	23.13	

DCR 2A - 11/15/2022: Wheat, silage, soft dough

Field name: DCR 2A

Crop: Wheat, silage, soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	191.86	22.62	356.82	662.95	607,500.00 *gal*
Application event totals		191.86	22.62	356.82	662.95	

DCR 2A - 06/13/2023: Corn, silage

Field name: DCR 2A

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	227.39	26.81	422.90	785.71	720,000.00 *gal*
MID	Surface water	6.34	0.00	0.00	22.63	7,051,414.40 *gal*
Application event totals		233.73	26.81	422.90	808.35	

DCR 2A - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.34	0.00	0.00	22.63	7,051,414.40 *gal*
Application event totals		6.34	0.00	0.00	22.63	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.34	0.00	0.00	22.63	7,051,414.40 *gal*
Application event totals		6.34	0.00	0.00	22.63	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.34	0.00	0.00	22.63	7,051,414.40 *gal*
Application event totals		6.34	0.00	0.00	22.63	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.34	0.00	0.00	22.63	7,051,414.40 *gal*
Application event totals		6.34	0.00	0.00	22.63	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.34	0.00	0.00	22.63	7,051,414.40 *gal*
Application event totals		6.34	0.00	0.00	22.63	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.34	0.00	0.00	22.63	7,051,414.40 *gal*
Application event totals		6.34	0.00	0.00	22.63	

DCR 2B - 11/15/2022: Wheat, silage, soft dough

DCR 2B - 11/15/2022: Wheat, silage, soft dough

Field name: DCR 2B

Crop: Wheat, silage, soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	252.00	13.97	197.09	897.19	881,250.00 *gal*
Application event totals		252.00	13.97	197.09	897.19	

DCR 2B - 06/14/2023: Corn, silage

Field name: DCR 2B

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.33	0.00	0.00	22.59	8,122,812.30 *gal*
Application event totals		6.33	0.00	0.00	22.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.33	0.00	0.00	22.59	8,122,812.30 *gal*
Application event totals		6.33	0.00	0.00	22.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	294.01	19.51	158.22	953.33	1,440,000.00 *gal*
MID	Surface water	6.33	0.00	0.00	22.59	8,122,812.30 *gal*
Application event totals		300.34	19.51	158.22	975.93	

DCR 2B - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.33	0.00	0.00	22.59	8,122,812.30 *gal*
Application event totals		6.33	0.00	0.00	22.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.33	0.00	0.00	22.59	8,122,812.30 *gal*
Application event totals		6.33	0.00	0.00	22.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	63.27	0.00	0.00	225.95	81,228,121.30 *gal*
Application event totals		63.27	0.00	0.00	225.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.33	0.00	0.00	22.59	8,122,812.30 *gal*
Application event totals		6.33	0.00	0.00	22.59	

DCR 2C - 11/18/2022: Wheat, silage, soft dough

Field name: DCR 2C

Crop: Wheat, silage, soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	153.97	18.15	286.34	531.99	337,500.00 *gal*
Application event totals		153.97	18.15	286.34	531.99	

DCR 2C - 06/15/2023: Corn, silage

Field name: DCR 2C

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.49	0.00	0.00	23.18	5,000,182.70 *gal*
Application event totals		6.49	0.00	0.00	23.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.49	0.00	0.00	23.18	5,000,182.70 *gal*
Application event totals		6.49	0.00	0.00	23.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.49	0.00	0.00	23.18	5,000,182.70 *gal*
Application event totals		6.49	0.00	0.00	23.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.49	0.00	0.00	23.18	5,000,182.70 *gal*
Application event totals		6.49	0.00	0.00	23.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	275.64	18.29	148.33	893.75	810,000.00 *gal*
MID	Surface water	6.49	0.00	0.00	23.18	5,000,182.70 *gal*
Application event totals		282.13	18.29	148.33	916.93	

DCR 2C - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.49	0.00	0.00	23.18	5,000,182.70 *gal*
Application event totals		6.49	0.00	0.00	23.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.49	0.00	0.00	23.18	5,000,182.70 *gal*
Application event totals		6.49	0.00	0.00	23.18	

DCR 2D - 11/18/2022: Wheat, silage, soft dough

Field name: DCR 2D

Crop: Wheat, silage, soft dough Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	252.92	14.02	197.81	900.48	855,000.00 *gal*
Application event totals		252.92	14.02	197.81	900.48	

DCR 2D - 06/16/2023: Corn, silage

Field name: DCR 2D

Crop: Corn, silage Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

DCR 2D - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.46	0.00	0.00	23.06	8,012,674.70 *gal*
Application event totals		6.46	0.00	0.00	23.06	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.46	0.00	0.00	23.06	8,012,674.70 *gal*
Application event totals		6.46	0.00	0.00	23.06	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	304.15	20.18	163.68	986.21	1,440,000.00 *gal*
MID	Surface water	6.46	0.00	0.00	23.06	8,012,674.70 *gal*
Application event totals		310.61	20.18	163.68	1,009.26	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.46	0.00	0.00	23.06	8,012,674.70 *gal*
Application event totals		6.46	0.00	0.00	23.06	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.46	0.00	0.00	23.06	8,012,674.70 *gal*
Application event totals		6.46	0.00	0.00	23.06	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.46	0.00	0.00	23.06	8,012,674.70 *gal*
Application event totals		6.46	0.00	0.00	23.06	

DCR 2D - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	6.46	0.00	0.00	23.06	8,012,674.70 *gal*
Application event totals		6.46	0.00	0.00	23.06	

OC 1 - 11/18/2022: Wheat, silage, soft dough

Field name: OC 1

Crop: Wheat, silage, soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	211.12	11.71	165.11	751.64	885,940.00 *gal*
Application event totals		211.12	11.71	165.11	751.64	

OC 1 - 07/03/2023: Corn, silage

Field name: OC 1

Crop: Corn, silage

Plant date: 07/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	11.47	0.00	0.00	40.97	17,674,155.00 *gal*
Application event totals		11.47	0.00	0.00	40.97	

OC 1 - 07/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	Light rain	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	7.04	0.00	0.00	25.15	10,847,578.00 *gal*
Application event totals		7.04	0.00	0.00	25.15	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	183.76	12.19	98.89	595.83	1,080,000.00 *gal*
MID	Surface water	5.06	0.00	0.00	18.08	7,800,870.00 *gal*
Application event totals		188.82	12.19	98.89	613.92	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	3.50	0.00	0.00	12.49	5,386,316.00 *gal*
Application event totals		3.50	0.00	0.00	12.49	

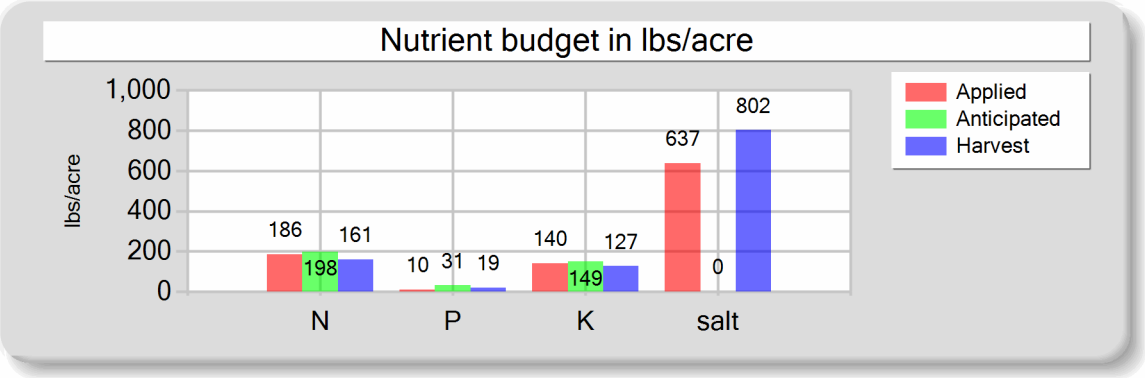
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID	Surface water	3.43	0.00	0.00	12.26	5,288,560.00 *gal*
Application event totals		3.43	0.00	0.00	12.26	

B. NUTRIENT BUDGET

CS 1 - 11/16/2022: Wheat, silage, soft dough

Field name: CS 1 Crop: Wheat, silage, soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	178.85	9.92	139.88	636.78
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.85	9.92	139.88	636.78
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	160.76	18.90	126.70	802.33
Nutrient balance	25.10	-8.98	13.19	-165.55
Applied to removed ratio	1.16	0.52	1.10	0.79

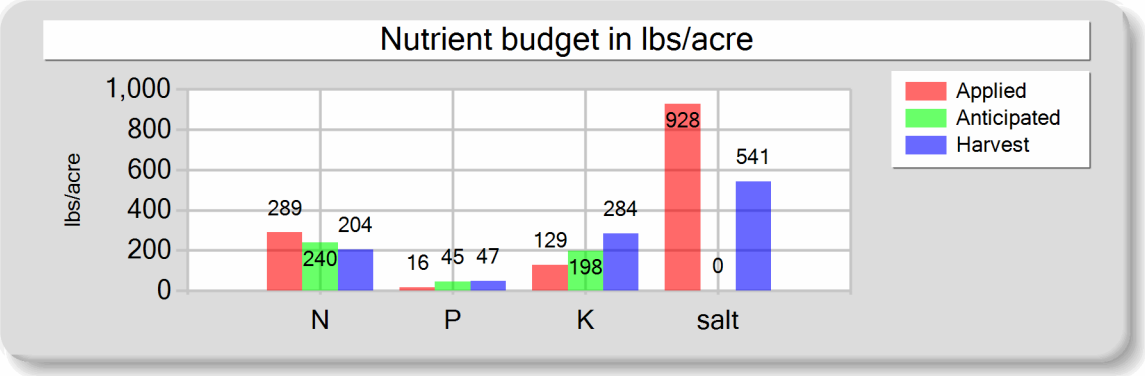
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,438,560.00	*gallons*
52.98	*acre-inches*
0.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

CS 1 - 06/21/2023: Corn, silage

Field name: CS 1 Crop: Corn, silage Plant date: 06/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	239.68	15.90	128.98	777.17
Fresh water	42.18	0.00	0.00	150.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	288.86	15.90	128.98	927.80
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	204.48	47.33	284.47	541.44
Nutrient balance	84.38	-31.43	-155.48	386.36
Applied to removed ratio	1.41	0.34	0.45	1.71

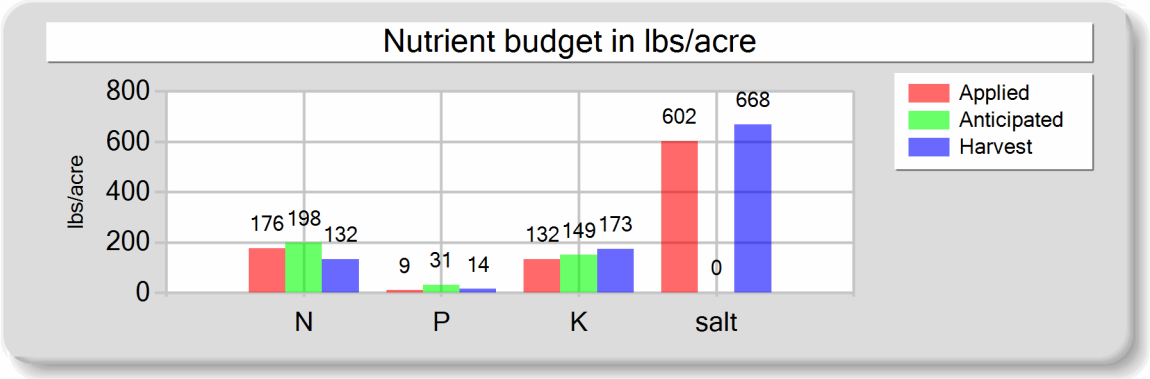
Fresh water applied	
124,543,814.50	*gallons*
4,586.53	*acre-inches*
66.47	*inches/acre*

Process wastewater applied	
2,700,000.00	*gallons*
99.43	*acre-inches*
1.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

CS 2 - 11/16/2022: Wheat, silage, soft dough

Field name: CS 2 Crop: Wheat, silage, soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	169.05	9.37	132.22	601.88
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	176.05	9.37	132.22	601.88
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	132.27	14.33	172.97	668.25
Nutrient balance	43.78	-4.95	-40.75	-66.37
Applied to removed ratio	1.33	0.65	0.76	0.90

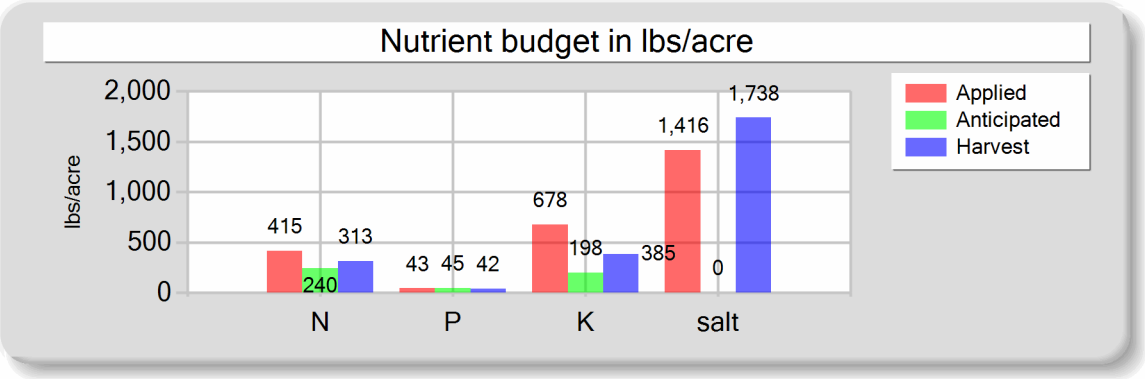
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,438,560.00	*gallons*
52.98	*acre-inches*
0.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

CS 2 - 06/08/2023: Corn, silage

Field name: CS 2 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	364.45	42.96	677.80	1,259.29
Fresh water	43.97	0.00	0.00	157.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	415.42	42.96	677.80	1,416.32
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	313.04	41.66	385.05	1,737.79
Nutrient balance	102.38	1.30	292.75	-321.47
Applied to removed ratio	1.33	1.03	1.76	0.82

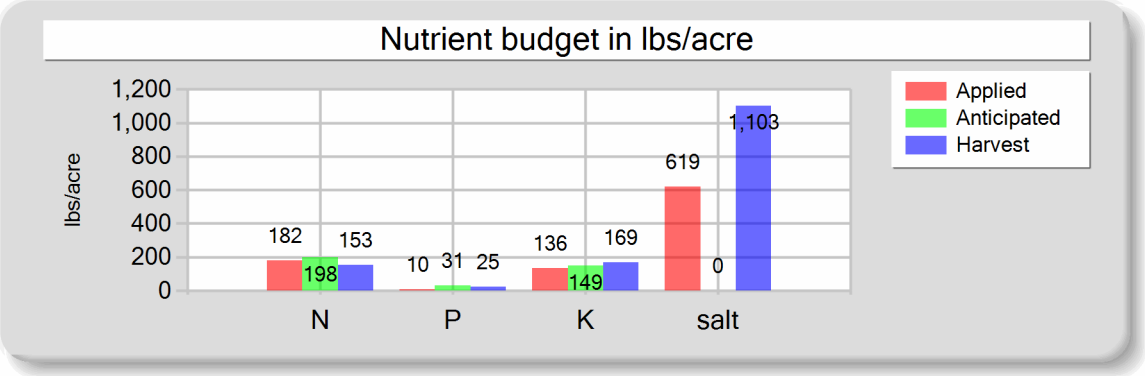
Fresh water applied	
137,361,487.20	*gallons*
5,058.56	*acre-inches*
69.30	*inches/acre*

Process wastewater applied	
3,240,000.00	*gallons*
119.32	*acre-inches*
1.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

CS 3 - 11/16/2022: Wheat, silage, soft dough

Field name: CS 3 Crop: Wheat, silage, soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	173.82	9.64	135.94	618.84
Fresh water	0.90	0.00	0.00	0.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	181.71	9.64	135.94	619.34
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	152.87	24.53	169.41	1,103.23
Nutrient balance	28.85	-14.89	-33.47	-483.89
Applied to removed ratio	1.19	0.39	0.80	0.56

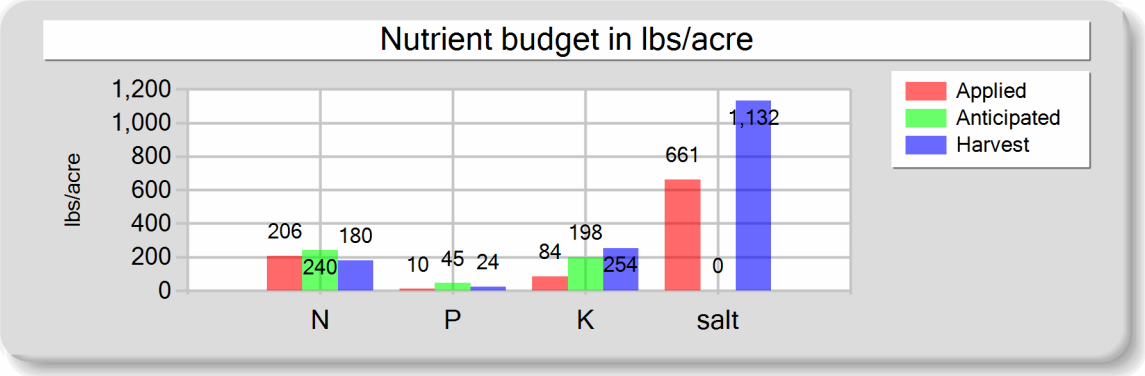
Fresh water applied	
7,128,000.00	*gallons*
262.50	*acre-inches*
3.70	*inches/acre*

Process wastewater applied	
1,438,560.00	*gallons*
52.98	*acre-inches*
0.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

CS 3 - 06/26/2023: Corn, silage

Field name: CS 3 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	155.29	10.30	83.57	503.52
Fresh water	44.03	0.00	0.00	157.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.32	10.30	83.57	660.78
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	180.39	24.15	254.24	1,131.74
Nutrient balance	25.94	-13.84	-170.68	-470.96
Applied to removed ratio	1.14	0.43	0.33	0.58

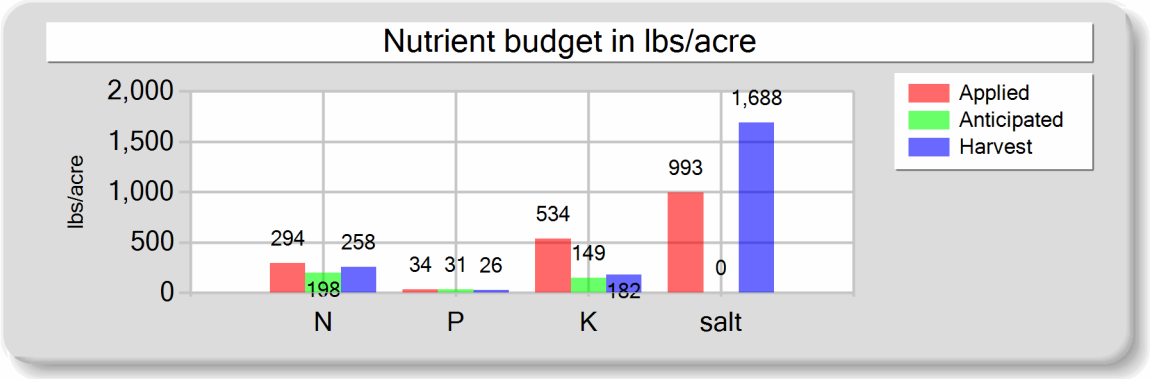
Fresh water applied	
133,800,912.00	*gallons*
4,927.43	*acre-inches*
69.40	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

DCR 1A - 11/15/2022: Wheat, silage, soft dough

Field name: DCR 1A Crop: Wheat, silage, soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	287.40	33.88	534.50	993.06
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.40	33.88	534.50	993.06
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	258.26	25.86	181.65	1,687.77
Nutrient balance	36.14	8.02	352.84	-694.71
Applied to removed ratio	1.14	1.31	2.94	0.59

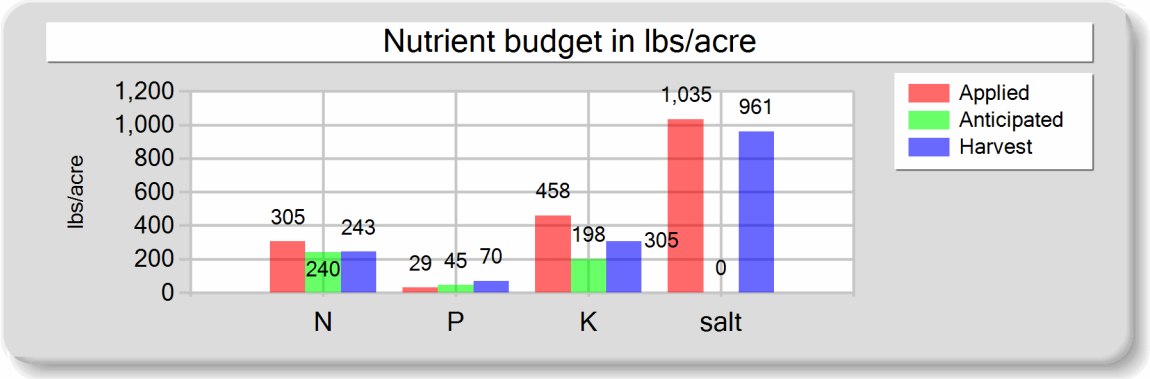
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,260,000.00 *gallons*
46.40 *acre-inches*
1.29 *inches/acre*

Total harvests for the crop
1 *harvests*

DCR 1A - 06/24/2023: Corn, silage

Field name: DCR 1A Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	246.34	29.04	458.14	851.19
Fresh water	51.33	0.00	0.00	183.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.67	29.04	458.14	1,034.50
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	243.07	69.76	304.78	960.86
Nutrient balance	61.60	-40.72	153.36	73.64
Applied to removed ratio	1.25	0.42	1.50	1.08

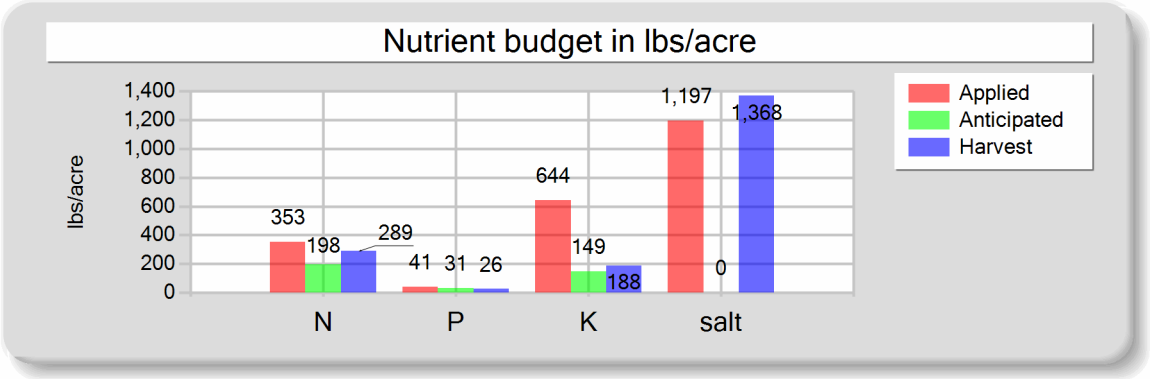
Fresh water applied	
79,080,764.00	*gallons*
2,912.28	*acre-inches*
80.90	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
1.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

DCR 1B - 11/15/2022: Wheat, silage, soft dough

Field name: DCR 1B Crop: Wheat, silage, soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	346.42	40.84	644.26	1,196.99
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	353.42	40.84	644.26	1,196.99
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	289.27	25.52	188.03	1,368.03
Nutrient balance	64.15	15.31	456.23	-171.04
Applied to removed ratio	1.22	1.60	3.43	0.87

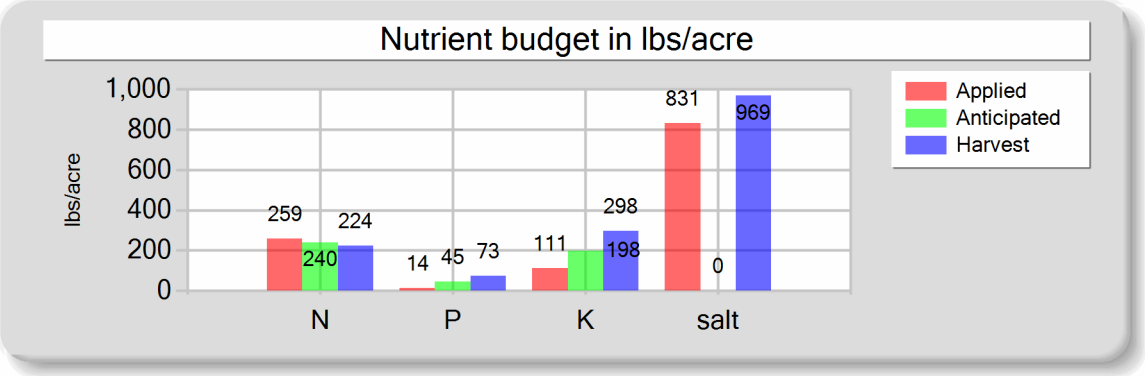
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
675,000.00	*gallons*
24.86	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

DCR 1B - 06/22/2023: Corn, silage

Field name: DCR 1B Crop: Corn, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	206.73	13.72	111.25	670.31
Fresh water	45.04	0.00	0.00	160.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	258.76	13.72	111.25	831.15
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	224.10	73.44	297.55	968.62
Nutrient balance	34.66	-59.73	-186.30	-137.47
Applied to removed ratio	1.15	0.19	0.37	0.86

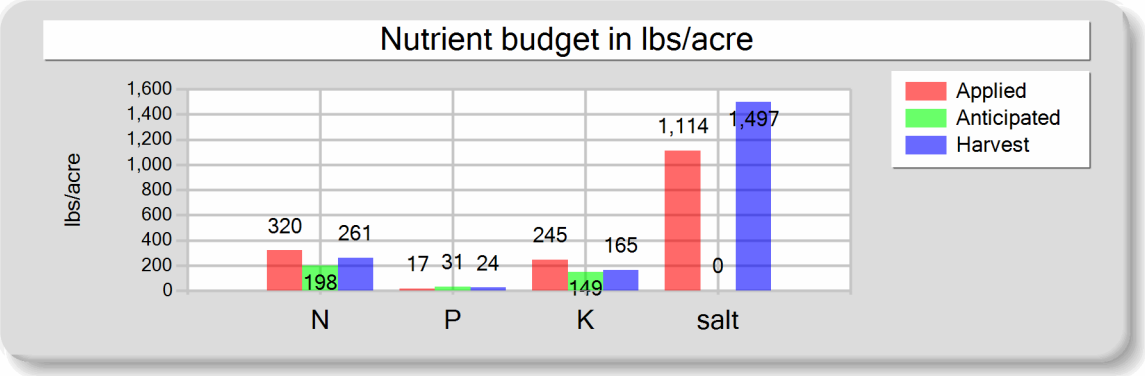
Fresh water applied	
30,838,533.67	*gallons*
1,135.68	*acre-inches*
70.98	*inches/acre*

Process wastewater applied	
540,000.00	*gallons*
19.89	*acre-inches*
1.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

DCR 1C - 11/15/2022: Wheat, silage, soft dough

Field name: DCR 1C | Crop: Wheat, silage, soft dough | Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	312.76	17.34	244.61	1,113.54
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	319.76	17.34	244.61	1,113.54
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	260.63	24.11	164.77	1,497.07
Nutrient balance	59.14	-6.76	79.84	-383.53
Applied to removed ratio	1.23	0.72	1.48	0.74

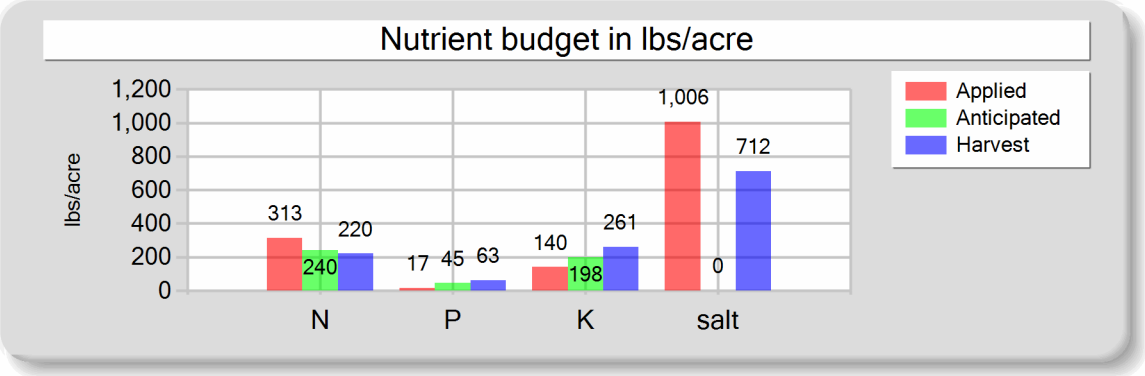
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,312,500.00	*gallons*
48.33	*acre-inches*
1.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

DCR 1C - 06/23/2023: Corn, silage

Field name: DCR 1C Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	260.71	17.30	140.30	845.34
Fresh water	45.04	0.00	0.00	160.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	312.74	17.30	140.30	1,006.18
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	220.33	62.62	260.82	711.85
Nutrient balance	92.41	-45.32	-120.53	294.33
Applied to removed ratio	1.42	0.28	0.54	1.41

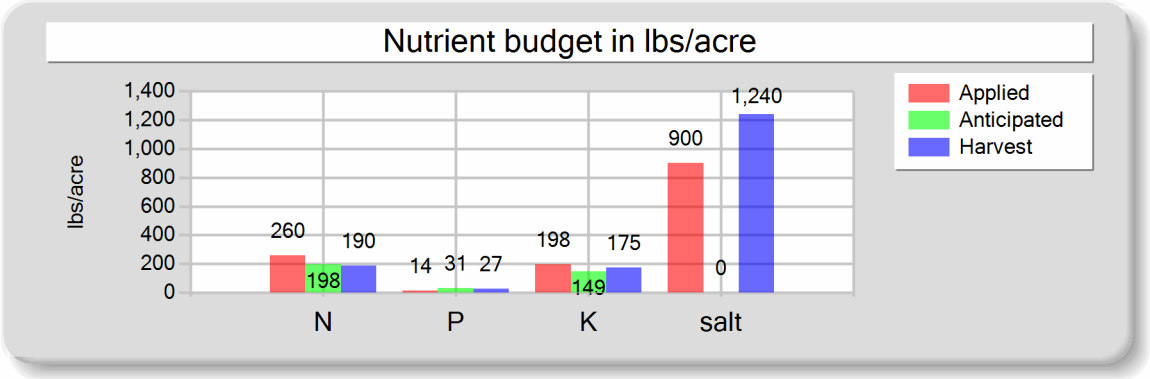
Fresh water applied	
69,386,699.90	*gallons*
2,555.28	*acre-inches*
70.98	*inches/acre*

Process wastewater applied	
1,532,250.00	*gallons*
56.43	*acre-inches*
1.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

DCR 1D - 11/15/2022: Wheat, silage, soft dough

Field name: DCR 1D Crop: Wheat, silage, soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	252.92	14.02	197.81	900.48
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	259.92	14.02	197.81	900.48
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	189.83	27.12	175.27	1,239.76
Nutrient balance	70.09	-13.09	22.54	-339.28
Applied to removed ratio	1.37	0.52	1.13	0.73

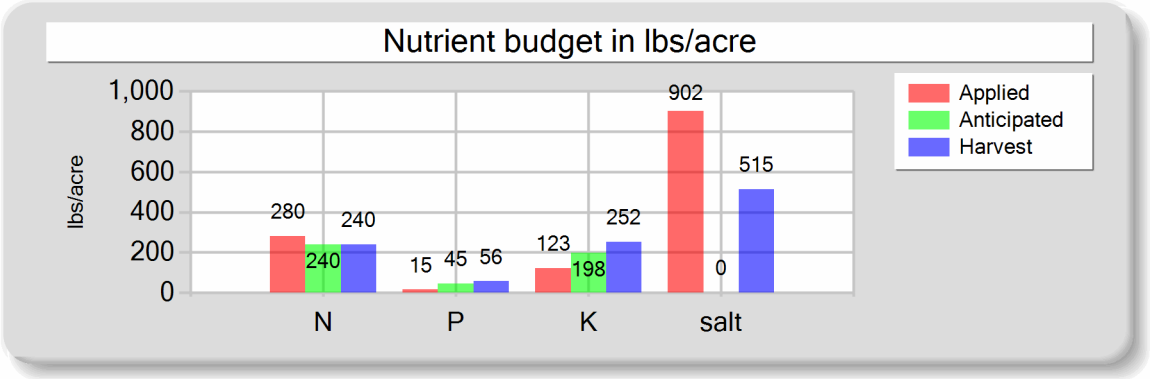
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
855,000.00 *gallons*
31.49 *acre-inches*
1.09 *inches/acre*

Total harvests for the crop
1 *harvests*

DCR 1D - 06/24/2023: Corn, silage

Field name: DCR 1D Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.11	15.13	122.76	739.65
Fresh water	45.33	0.00	0.00	161.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	280.44	15.13	122.76	901.55
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	240.26	56.39	252.11	514.64
Nutrient balance	40.19	-41.26	-129.35	386.92
Applied to removed ratio	1.17	0.27	0.49	1.75

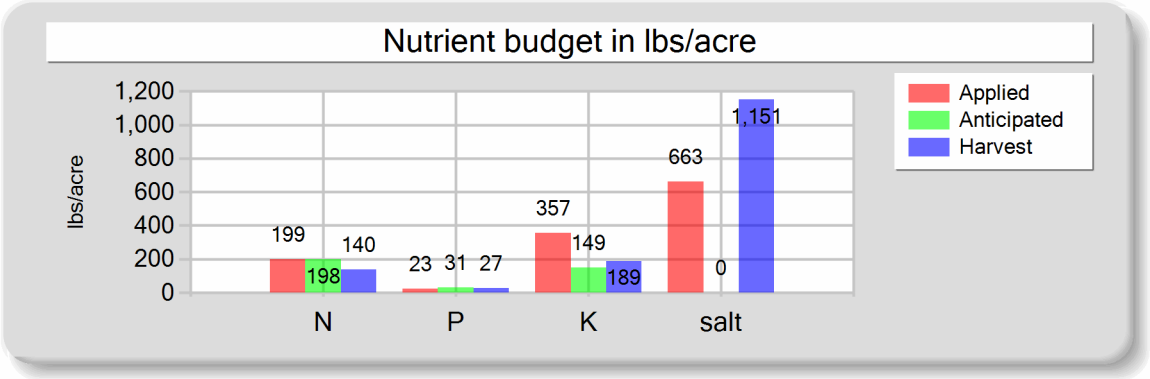
Fresh water applied	
56,262,075.80	*gallons*
2,071.94	*acre-inches*
71.45	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
1.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

DCR 2A - 11/15/2022: Wheat, silage, soft dough

Field name: DCR 2A Crop: Wheat, silage, soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	191.86	22.62	356.82	662.95
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.86	22.62	356.82	662.95
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	139.67	27.38	188.84	1,150.91
Nutrient balance	59.19	-4.76	167.98	-487.96
Applied to removed ratio	1.42	0.83	1.89	0.58

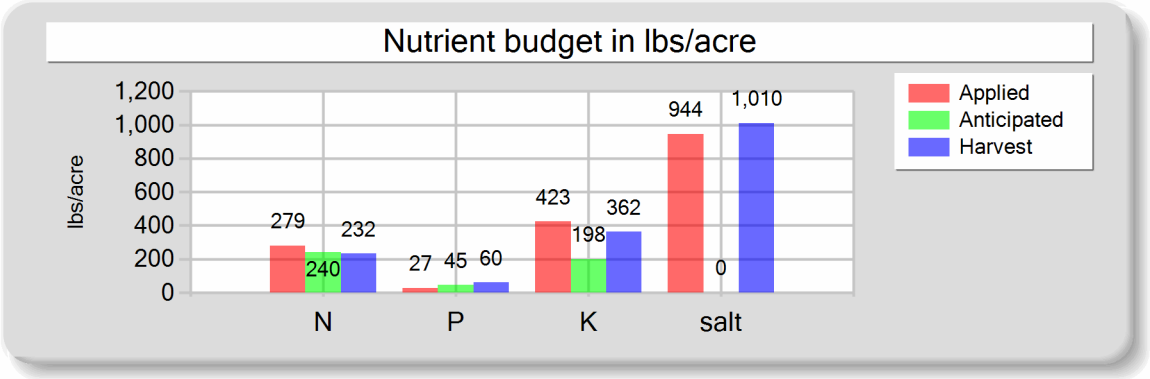
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
607,500.00 *gallons*
22.37 *acre-inches*
0.86 *inches/acre*

Total harvests for the crop
1 *harvests*

DCR 2A - 06/13/2023: Corn, silage

Field name: DCR 2A Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	227.39	26.81	422.90	785.71
Fresh water	44.36	0.00	0.00	158.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	278.75	26.81	422.90	944.14
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	231.76	59.95	361.83	1,009.83
Nutrient balance	46.99	-33.14	61.07	-65.69
Applied to removed ratio	1.20	0.45	1.17	0.93

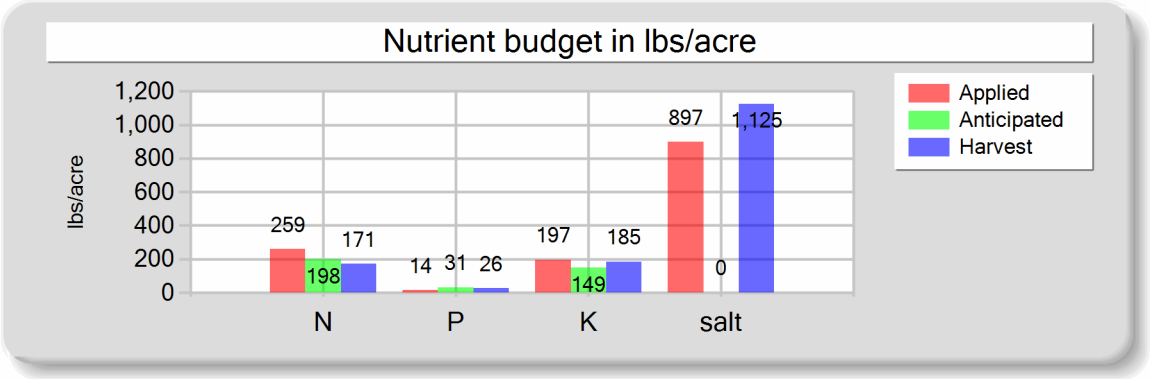
Fresh water applied	
49,359,900.80	*gallons*
1,817.76	*acre-inches*
69.91	*inches/acre*

Process wastewater applied	
720,000.00	*gallons*
26.52	*acre-inches*
1.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

DCR 2B - 11/15/2022: Wheat, silage, soft dough

Field name: DCR 2B | Crop: Wheat, silage, soft dough | Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	252.00	13.97	197.09	897.19
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	259.00	13.97	197.09	897.19
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	170.95	25.66	185.45	1,125.37
Nutrient balance	88.05	-11.68	11.64	-228.18
Applied to removed ratio	1.52	0.54	1.06	0.80

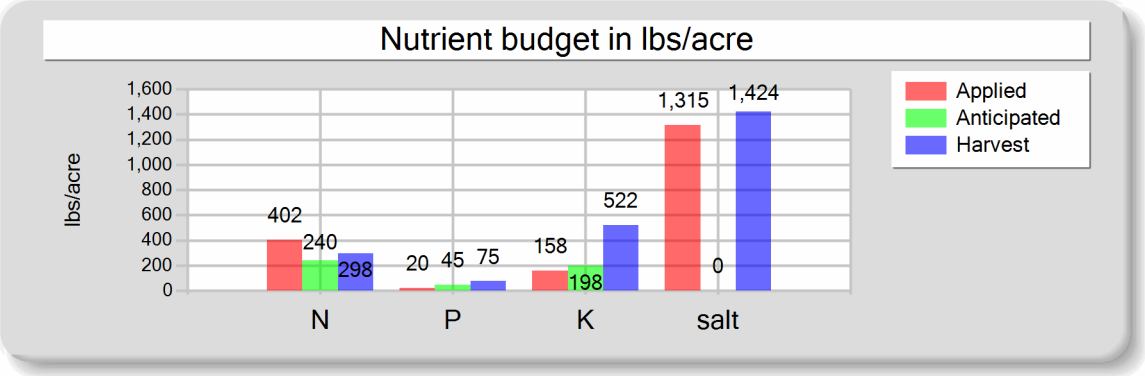
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
881,250.00	*gallons*
32.45	*acre-inches*
1.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

DCR 2B - 06/14/2023: Corn, silage

Field name: DCR 2B Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	294.01	19.51	158.22	953.33
Fresh water	101.23	0.00	0.00	361.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	402.24	19.51	158.22	1,314.85
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	297.63	74.81	521.52	1,424.35
Nutrient balance	104.61	-55.30	-363.30	-109.50
Applied to removed ratio	1.35	0.26	0.30	0.92

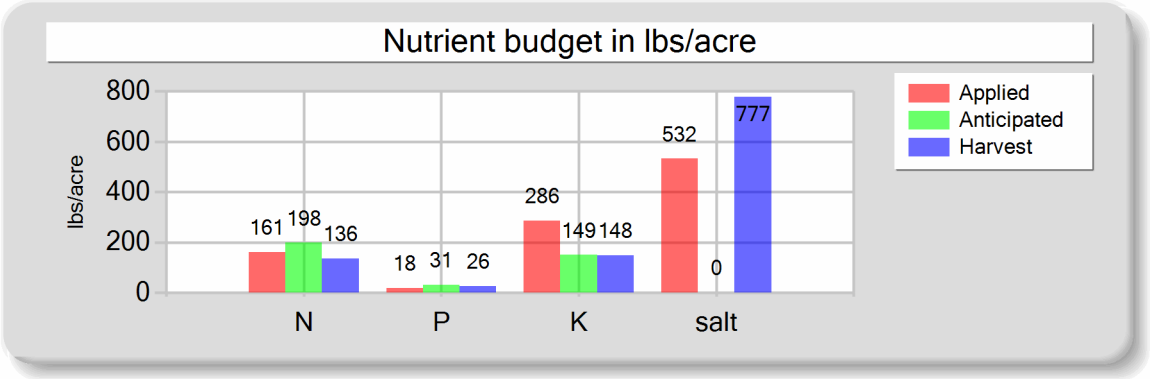
Fresh water applied	
129,964,995.10	*gallons*
4,786.17	*acre-inches*
159.54	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

DCR 2C - 11/18/2022: Wheat, silage, soft dough

Field name: DCR 2C Crop: Wheat, silage, soft dough Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	153.97	18.15	286.34	531.99
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	160.97	18.15	286.34	531.99
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	135.91	26.32	148.21	776.88
Nutrient balance	25.06	-8.17	138.13	-244.89
Applied to removed ratio	1.18	0.69	1.93	0.68

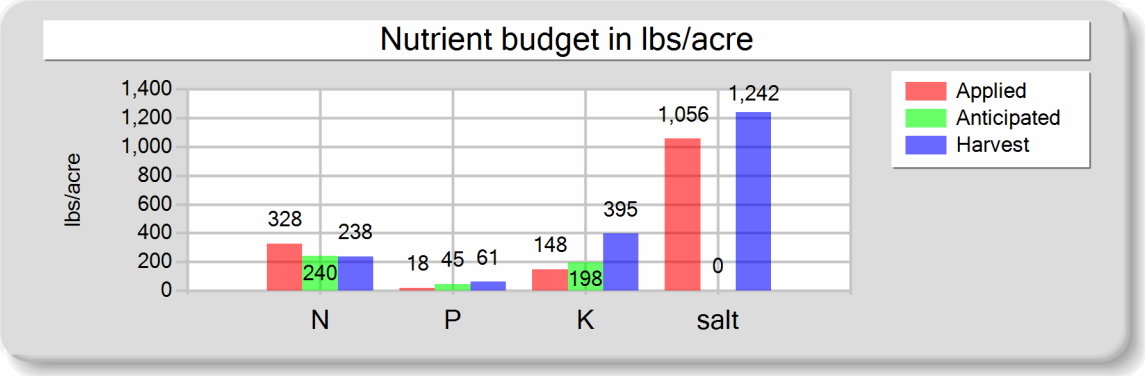
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
337,500.00	*gallons*
12.43	*acre-inches*
0.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

DCR 2C - 06/15/2023: Corn, silage

Field name: DCR 2C Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.64	18.29	148.33	893.75
Fresh water	45.44	0.00	0.00	162.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	328.07	18.29	148.33	1,056.02
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	238.48	60.85	395.28	1,241.89
Nutrient balance	89.59	-42.57	-246.94	-185.87
Applied to removed ratio	1.38	0.30	0.38	0.85

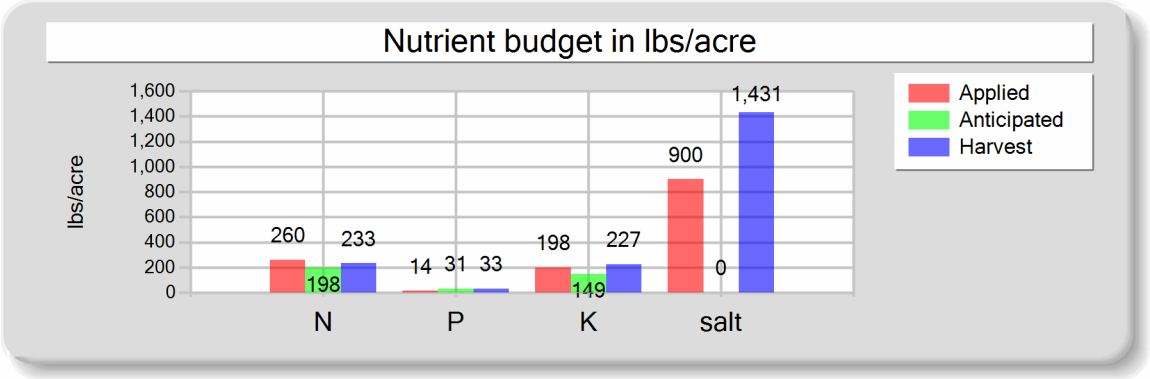
Fresh water applied	
35,001,278.90	*gallons*
1,288.98	*acre-inches*
71.61	*inches/acre*

Process wastewater applied	
810,000.00	*gallons*
29.83	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

DCR 2D - 11/18/2022: Wheat, silage, soft dough

Field name: DCR 2D | Crop: Wheat, silage, soft dough | Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	252.92	14.02	197.81	900.48
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	259.92	14.02	197.81	900.48
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	232.79	33.30	227.09	1,431.15
Nutrient balance	27.14	-19.27	-29.28	-530.67
Applied to removed ratio	1.12	0.42	0.87	0.63

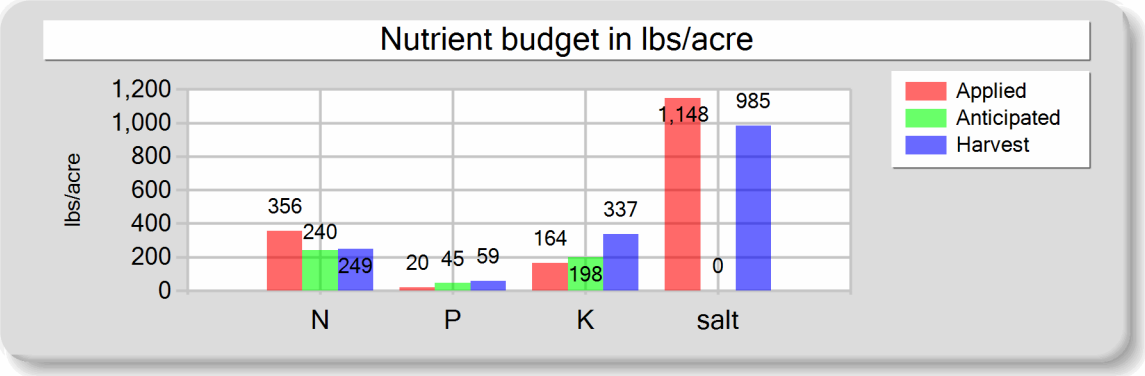
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
855,000.00 *gallons*
31.49 *acre-inches*
1.09 *inches/acre*

Total harvests for the crop
1 *harvests*

DCR 2D - 06/16/2023: Corn, silage

Field name: DCR 2D Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	304.15	20.18	163.68	986.21
Fresh water	45.19	0.00	0.00	161.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	356.34	20.18	163.68	1,147.61
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	248.65	58.89	336.98	984.64
Nutrient balance	107.70	-38.71	-173.30	162.96
Applied to removed ratio	1.43	0.34	0.49	1.17

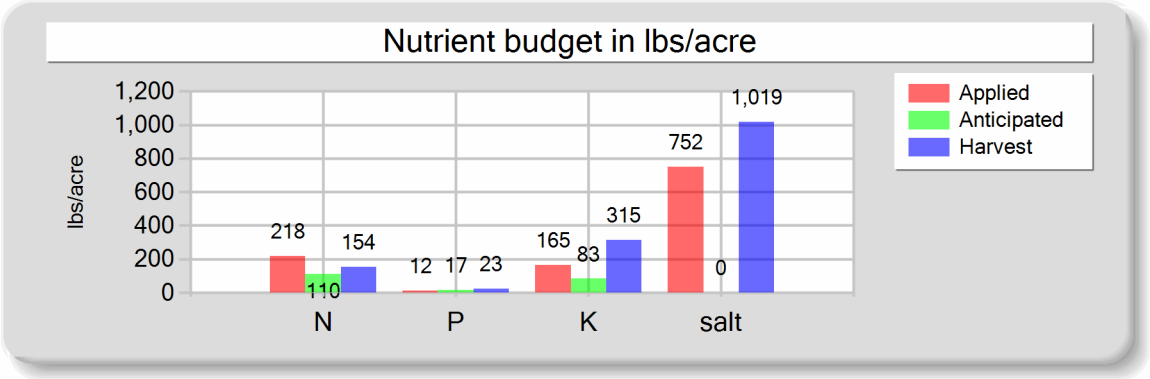
Fresh water applied	
56,088,722.90	*gallons*
2,065.56	*acre-inches*
71.23	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

OC 1 - 11/18/2022: Wheat, silage, soft dough

Field name: OC 1 Crop: Wheat, silage, soft dough Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	211.12	11.71	165.11	751.64
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.12	11.71	165.11	751.64
Anticipated crop nutrient removal	110.00	17.00	83.00	0.00
Actual crop nutrient removal	154.28	22.97	314.80	1,018.92
Nutrient balance	63.84	-11.27	-149.68	-267.28
Applied to removed ratio	1.41	0.51	0.52	0.74

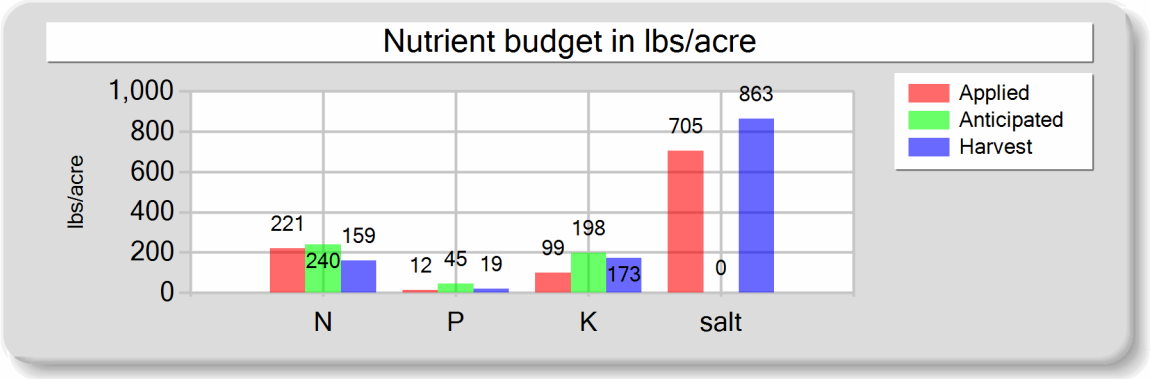
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
885,940.00	*gallons*
32.63	*acre-inches*
0.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

OC 1 - 07/03/2023: Corn, silage

Field name: OC 1 Crop: Corn, silage Plant date: 07/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.76	12.19	98.89	595.83
Fresh water	30.50	0.00	0.00	108.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.26	12.19	98.89	704.78
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	158.76	18.63	173.37	863.08
Nutrient balance	62.50	-6.43	-74.48	-158.31
Applied to removed ratio	1.39	0.65	0.57	0.82

Fresh water applied	
46,997,479.00	*gallons*
1,730.76	*acre-inches*
48.08	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
1.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Manure

Sample and source description: Separator Manure

Sample date: 04/13/2021 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	2,200.00	6,500.00							
DL	1,000.00	1,000.00	1,000.00							

Seperator Material

Sample and source description: Seperator Material

Sample date: 11/23/2021 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	1,700.00	7,900.00							
DL	1,000.00	1,000.00	1,000.00							

Corral Material

Sample and source description: Corral Material

Sample date: 03/15/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	4,700.00	4,700.00						0.00	
DL	1,000.00	1,000.00	1,000.00						1,000.00	

Corral Manure

Sample and source description: Corral Manure

Sample date: 10/04/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 16.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,100.00	6,800.00	42,000.00							
DL	1,000.00	1,000.00	1,000.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon

Sample and source description: Lagoon

Sample date: 05/17/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	703.00	378.00		325.00	57.00	804.00								6.67	3,660
DL	0.50	0.50		0.50	0.50	0.50								1.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 05/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	650.00	316.00		334.00	116.00	1,830.00								5,040.00	3,400
DL	0.50	0.50		0.50	0.50	2.00								1.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 10/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	507.00	280.00		227.00	48.70	395.00	81.50	64.20	132.00	2,040.00	2.00	222.00	171.00	4,300.00	2,380
DL	0.50	0.50		0.50	0.50	2.00	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

C. FRESH WATER ANALYSES

CSR Well

CSR Well

Sample description: CSR Well

Sample date: 03/17/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.07	0.50	0.57								1.00	
DL	0.50	0.50	0.50								1.00	

DCR Well

DCR Well

Sample description: DCR Well

Sample date: 03/17/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.58	0.50	7.08								1.00	
DL	0.50	0.50	0.50								1.00	

MID

MID

MID Irrigation Water

Sample description: MID Irrigation Water

Sample date: 05/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.80	0.50	0.50								1.00	10
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

CS 1

CS 1

Sample and source description: CS 1

Sample date: 05/16/2018 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			77.00				
DL			1.00				

CS 2

CS 2

Sample and source description: CS 2

Sample date: 05/23/2018 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			70.00				
DL			1.00				

CS 3

CS 3

CS 3

Sample and source description: CS 3

Sample date: 05/16/2018 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			96.00				
DL			1.00				

DCR 1A

DC 1A

Sample and source description: DC 1A

Sample date: 10/04/2017 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			99.00				
DL			1.00				

DCR 1B

DC 1B

Sample and source description: DC 1B

Sample date: 10/04/2017 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			132.00				
DL			1.00				

DCR 1C

DCR 1C

DC 1C

Sample and source description: DC 1C

Sample date: 05/16/2018 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			78.00				
DL			1.00				

DCR 1D

DC 1D

Sample and source description: DC 1D

Sample date: 05/16/2018 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			116.00				
DL			1.00				

DCR 2A

DC 2A

Sample and source description: DC 2A

Sample date: 09/06/2017 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			93.00				
DL			1.00				

DCR 2B

DCR 2B

DC 2B

Sample and source description: DC 2B

Sample date: 09/06/2017 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			128.00				
DL			1.00				

DCR 2C

DC 2C

Sample and source description: DC 2C

Sample date: 09/20/2017 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			156.00				
DL			1.00				

DCR 2D

DC 2D

Sample and source description: DC 2D

Sample date: 09/20/2017 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			177.00				
DL			1.00				

E. PLANT TISSUE ANALYSES

CS 1 - 11/16/2022: Wheat, silage, soft dough

CS 1 - 11/16/2022: Wheat, silage, soft dough

Wheat Silage

Sample and source description: Wheat Silage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 54.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,740.00	910.00	6,100.00		8.49
DL	0.50	0.50	0.50		0.01

CS 1 - 06/21/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,320.00	1,000.00	6,010.00	100.00	3.10
DL	100.00	100.00	100.00	100.00	0.01

CS 2 - 11/16/2022: Wheat, silage, soft dough

Wheat Silage

Sample and source description: Wheat Silage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,370.00	690.00	8,330.00		8.89
DL	100.00	100.00	100.00		0.01

CS 2 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,260.00	700.00	6,470.00	100.00	8.00
DL	100.00	100.00	100.00	100.00	0.01

CS 3 - 11/16/2022: Wheat, silage, soft dough

Wheat Silage

Sample and source description: Wheat Silage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,360.00	860.00	5,940.00	100.00	10.10
DL	100.00	100.00	100.00	100.00	0.01

CS 3 - 06/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,810.00	510.00	5,370.00	100.00	7.20
DL	100.00	100.00	100.00	100.00	0.01

DCR 1A - 11/15/2022: Wheat, silage, soft dough

Wheat Silage

Sample and source description: Wheat Silage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,990.00	800.00	5,620.00	100.00	12.20
DL	100.00	100.00	100.00	100.00	0.01

DCR 1A - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,530.00	1,300.00	5,680.00	100.00	4.70
DL	100.00	100.00	100.00	100.00	0.01

DCR 1B - 11/15/2022: Wheat, silage, soft dough

Wheat Silage

Sample and source description: Wheat Silage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 45.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	900.00	6,630.00	100.00	8.90
DL	100.00	100.00	100.00	100.00	0.01

DCR 1B - 06/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,760.00	1,560.00	6,320.00		5.40
DL	100.00	100.00	100.00		0.01

DCR 1C - 11/15/2022: Wheat, silage, soft dough

Wheat Silage

Sample and source description: Wheat Silage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 50.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,190.00	850.00	5,810.00	100.00	10.60
DL	100.00	100.00	100.00	100.00	0.01

DCR 1C - 06/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,680.00	1,330.00	5,540.00		4.20
DL	100.00	100.00	100.00		0.01

DCR 1D - 11/15/2022: Wheat, silage, soft dough

Wheat Silage

Sample and source description: Wheat Silage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 53.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,650.00	950.00	6,140.00	100.00	9.34
DL	100.00	100.00	100.00	100.00	0.01

DCR 1D - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,070.00	1,190.00	5,320.00		3.00
DL	100.00	100.00	100.00		0.01

DCR 2A - 11/15/2022: Wheat, silage, soft dough

Wheat Silage

Sample and source description: Wheat Silage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,000.00	980.00	6,760.00	100.00	10.30
DL	100.00	100.00	100.00	100.00	0.01

DCR 2A - 06/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,330.00	1,120.00	6,760.00	100.00	5.27
DL	100.00	100.00	100.00	100.00	0.01

DCR 2B - 11/15/2022: Wheat, silage, soft dough

Wheat Silage

Sample and source description: Wheat Silage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,130.00	920.00	6,650.00	100.00	9.54
DL	100.00	100.00	100.00	100.00	0.01

DCR 2B - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 52.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,570.00	1,400.00	9,760.00	100.00	5.60
DL	100.00	100.00	100.00	100.00	0.01

DCR 2C - 11/18/2022: Wheat, silage, soft dough

Wheat Silage

Sample and source description: Wheat Silage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,750.00	920.00	5,180.00	100.00	7.78
DL	100.00	100.00	100.00	100.00	0.01

DCR 2C - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,350.00	1,110.00	7,210.00	100.00	5.93
DL	100.00	100.00	100.00	100.00	0.01

DCR 2D - 11/18/2022: Wheat, silage, soft dough

Wheat Silage

Sample and source description: Wheat Silage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 53.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,180.00	1,170.00	7,980.00	100.00	10.70
DL	100.00	100.00	100.00	100.00	0.01

DCR 2D - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,560.00	1,080.00	6,180.00	100.00	5.03
DL	100.00	100.00	100.00	100.00	0.01

OC 1 - 11/18/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,440.00	810.00	5,330.00		9.98
DL	100.00	100.00	100.00		0.01

OC 1 - 07/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,540.00	650.00	6,050.00		7.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

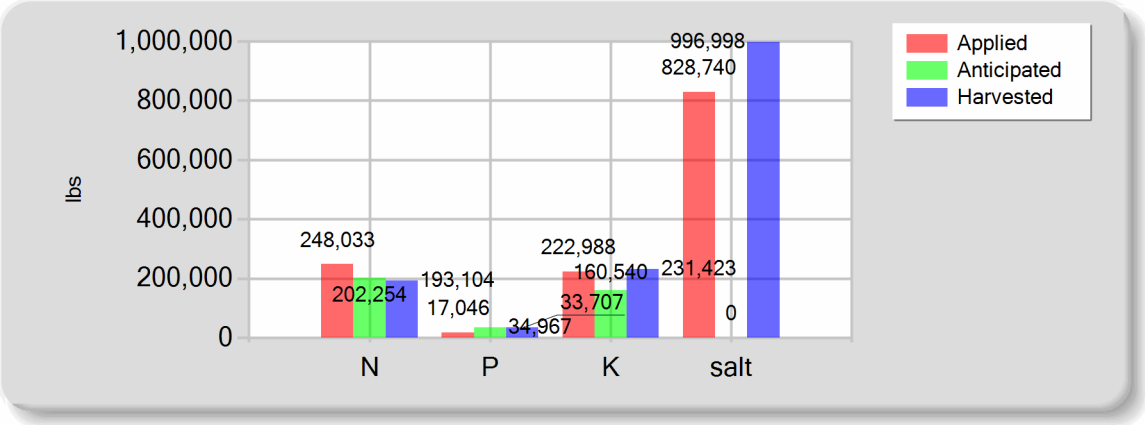
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

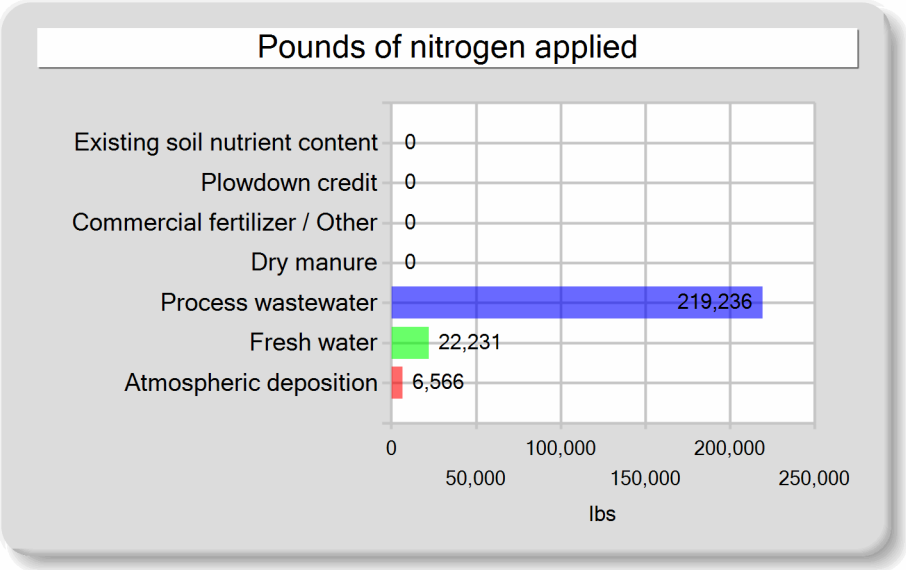
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

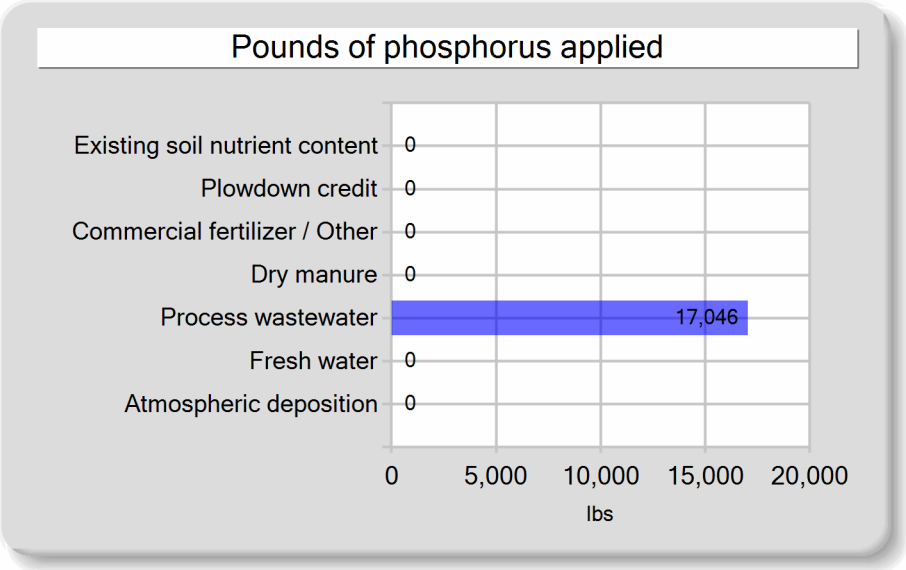
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	219,235.93	17,046.25	222,987.63	749,536.29
Fresh water	22,230.66	0.00	0.00	79,203.59
Atmospheric deposition	6,566.00	0.00	0.00	0.00
Total nutrients applied	248,032.59	17,046.25	222,987.63	828,739.89
Anticipated crop nutrient removal	202,254.00	34,966.80	160,540.20	0.00
Actual crop nutrient removal	193,103.54	33,706.82	231,422.87	996,998.34
Nutrient balance	54,929.06	-16,660.57	-8,435.24	-168,258.46
Applied to removed ratio	1.28	0.51	0.96	0.83

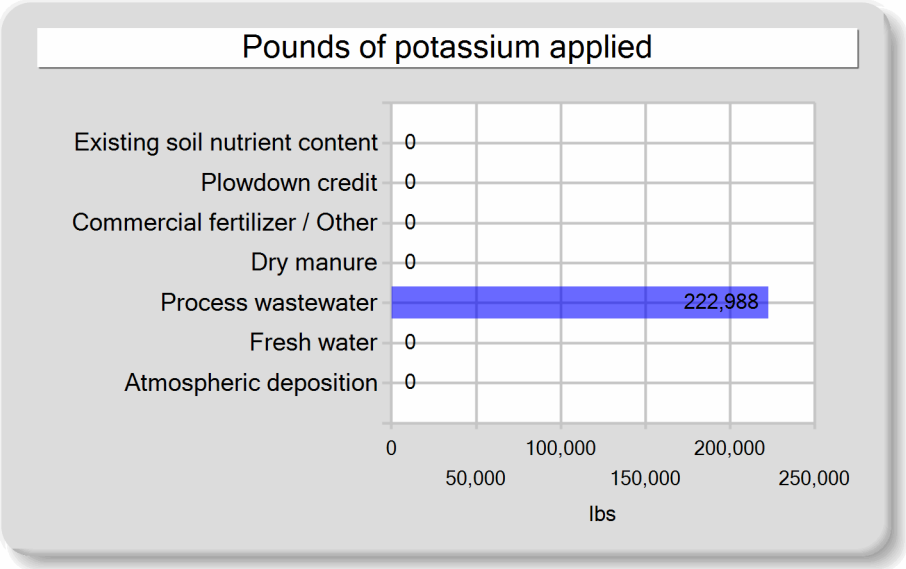
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

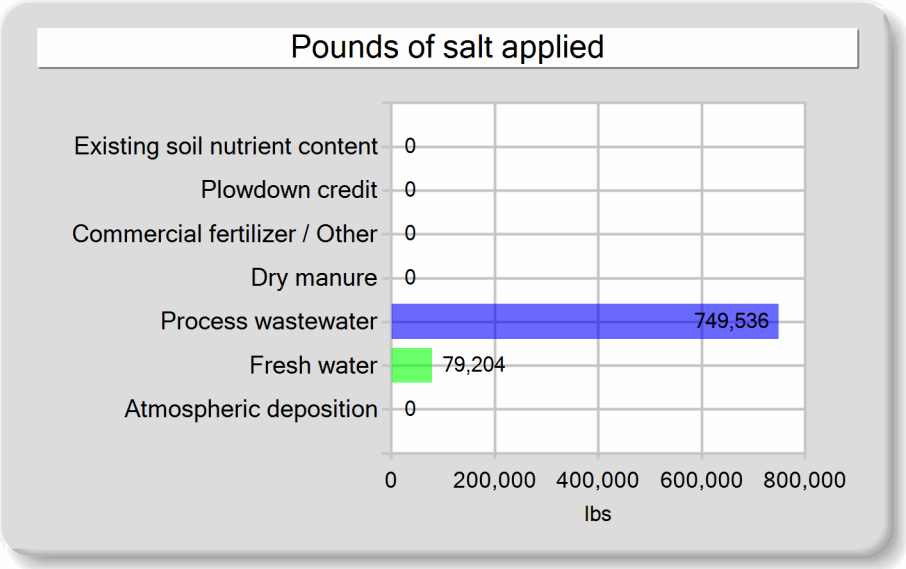


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Robert & Victoria Strickland	Matt Strickland
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Matt Strickland

Name of Dairy Facility: Double Creek Dairy

Facility Address:

1320 S Arboleda DR	Merced	Merced	95340
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Matt Strickland (209) 769-2813
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Seth & Machelle Rossow Farms

Address of Hauling Company/Person:

7877 E Sandy Mush RD	Merced	CA	95341
Number and Street	City	State	Zip Code

Contact Person: Seth Rossow (209) 617-7755
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Seth & Mechelle Rossow Farms (209) 617-7755
Name Phone Number

7877 E Sandy Mush RD	Merced	CA	95341
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

7877 E Sandy Mush	Merced	95341
Address	City	Zip Code

	Merced
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 05/15/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 640.00 tons

Manure Solids Content: 83.7 %

Method used to determine amount of manure:

Scale

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator Signature Date

Hauler Signature Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Matt Strickland

Name of Dairy Facility: Double Creek Dairy

Facility Address:

1320 S Arboleda DR	Merced	Merced	95340
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Matt Strickland (209) 769-2813
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Seth & Machelle Rossow Farms

Address of Hauling Company/Person:

7877 E Sandy Mush RD	Merced	CA	95341
Number and Street	City	State	Zip Code

Contact Person: Seth Rossow (209) 617-7755
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Seth & Mechelle Rossow Farms (209) 617-7755
Name Phone Number

7877 E Sandy Mush RD	Merced	CA	95341
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

7877 E Sandy Mush	Merced	95341
Address	City	Zip Code

	Merced
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 10/20/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 2,790.00 tons

Manure Solids Content: 83.7 %

Method used to determine amount of manure:

Scale weights

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator Signature Date

Hauler Signature Date

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Double Creek Dairy
PO Box 2128
Merced, CA 95344

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 4/26/24 10:00 AM
SUBMITTED BY: Matt
ANALYZED DATE: 4/26/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature Above Qualifying at 17.7

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well #2	T4117027E	Electrical Conductivity (EC)	EPA 120.1 and	0.322	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.67	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	206	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	104 (T4117027D)	100 (T4117027D)	3.53
Electrical Conductivity (EC)	97.5			1.47
Nitrate Nitrogen (NO3-N)	99.1	102 (T4116005A)	101 (T4116005A)	1.32

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Double Creek Dairy
PO Box 2128
Merced, CA 95344

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 4/26/24 10:00 AM
SUBMITTED BY: Matt
ANALYZED DATE: 4/26/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature Above Qualifying at 17.4

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well #3	T4117027D	Electrical Conductivity (EC)	EPA 120.1 and	0.322	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.81	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	206	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	104 (T4117027D)	100 (T4117027D)	3.53
Electrical Conductivity (EC)	97.5			1.47
Nitrate Nitrogen (NO3-N)	99.1	102 (T4116005A)	101 (T4116005A)	1.32

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: DOUBLE CREEK DAIRY
Mailing address: P.O. BOX 2128
(not facility address) MERCED CA 95344
Phone#: 209-769-2813

Authorized Copy Released To: F+R

Sampled By: MATT
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	4/26/24	8:30AM	DC2C	WHEAT SILAGE							F18+
2	4/26/24	9:30AM	DC2A	"							F18+
3	4/25/24	5:00PM	DC2D	"							F18+
4	4/26/24	9:00AM	WELL #3	DOMESTIC	17.4		X				
5	4/26/24	8:45AM	WELL #2	DOMESTIC	17.7		X				
6	4/26/24	9:15AM	LAGOON					X			
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	[signature]			4-26-24		
2nd	[signature]	DENELE ANALYTICAL, INC.	4-26-24 10:40			T4117027

Required to be submitted to any agency? (YES / NO)